

Netflix Content Analysis & Visualization Dashboard

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Company: Horizon TechX

Domain: AI & Data Science

Task: Data Visualization Dashboard

Dataset: Netflix Movies and TV Shows Dataset

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Google Colab

Date: May 2026

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1. Introduction

1.1 Project Overview

This project was completed as part of the Horizon TechX AI & Data Science Internship Program. The primary objective of this project is to perform a comprehensive analysis and create professional data visualizations on Netflix's content library to extract meaningful insights about content strategy, geographical distribution, genre preferences, and temporal addition patterns.

Data visualization is a critical component of modern data science that transforms raw data into compelling visual narratives. This project demonstrates industry-standard visualization techniques using Python and its powerful libraries to create interactive and static dashboards for strategic business decision-making.

1.2 Objectives

- To clean and preprocess Netflix content data for analysis.
- To perform comprehensive statistical analysis on content distribution.
- To identify content patterns across different categories and regions.
- To analyze temporal trends in Netflix's content addition strategy.
- To examine genre preferences and rating distributions.
- To generate professional data visualizations for business insights.
- To create a comprehensive dashboard for strategic decision-making.
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1.3 Scope

This analysis covers Netflix's complete content catalog from 2008 to 2021, spanning 87 countries and examining 8,797 titles across movies and TV shows. The analysis includes content distribution, geographical patterns, genre analysis, rating classifications, duration patterns, and director insights.

2. Dataset Description

2.1 Dataset Source

Platform: Kaggle

Dataset Name: Netflix Movies and TV Shows

URL: <https://www.kaggle.com/shivamb/netflix-shows>

Format: CSV (Comma Separated Values)

2.2 Dataset Overview

Attribute	Value
Total Records (Raw)	8,807
Total Records (Cleaned)	8,797
Total Columns	12 (original) → 20 (after feature engineering)
Time Period	01 January 2008 — 25 September 2021
Geographic Coverage	87 Countries

2.3 Column Description

Column Name	Data Type	Description
show_id	Object	Unique identifier for each title
type	Object	Content type (Movie/TV Show)
title	Object	Name of the movie or TV show
director	Object	Director(s) of the content
cast	Object	Main cast members
country	Object	Country/countries of production

2.4 Engineered Features

After preprocessing, the following new columns were added:

New Column	Description
year_added	Year extracted from date_added
month_added	Month number extracted from date_added
month_name	Month name extracted from date_added
day_added	Day extracted from date_added
duration_numeric	Numeric duration for movies (minutes) / seasons for TV shows
primary_genre	First genre from the listed_in field
primary_country	First country from the country field
genre_count	Number of genres per title

3. Data Preprocessing & Cleaning

3.1 Missing Values Analysis

Before cleaning, the dataset contained the following missing values:

Column	Missing Values	Percentage
show_id	0	0.00%
type	0	0.00%
title	0	0.00%
director	2,634	29.91%
cast	825	9.37%
country	831	9.44%
date_added	10	0.11%
release_year	0	0.00%
rating	4	0.05%
duration	3	0.03%
listed_in	0	0.00%
description	0	0.00%

3.2 Cleaning Steps Performed

The following cleaning operations were performed on the raw dataset:

Step	Action	Reason
1	Filled missing director with 'Unknown'	Many titles lack credited directors
2	Filled missing cast with 'Not Available'	Cast information not always available
3	Filled missing country with 'Unknown'	Some content has unclear origin
4	Removed rows with missing date_added	Essential for temporal analysis
5	Filled missing rating with mode value	Required for rating analysis
6	Filled missing duration with 'Unknown'	Maintain complete dataset
7	Converted date_added to DateTime format	Required for time series analysis
8	Extracted year, month, day components	Enable temporal pattern analysis
9	Created duration_numeric column	Standardize duration for analysis
10	Extracted primary_genre and primary_country	Simplify categorical analysis

3.3 Cleaning Results

Metric	Before Cleaning	After Cleaning
Total Rows	8,807	8,797
Total Columns	12	20
Rows Removed	—	10
Missing Values	4,307	0
New Columns Added	—	8

4. Content Performance Overview

4.1 Key Performance Indicators (KPIs)

KPI	Value
 Total Titles Analyzed	8,797
 Total Movies	6,131 (69.7%)
 Total TV Shows	2,666 (30.3%)
 Countries Represented	87
 Unique Genres	36
 Total Directors	4,529
 Rating Categories	17
 Data Period	Jan 2008 — Sep 2021
 Peak Addition Year	2019 (1,424 titles)
 Most Popular Genre	International Movies
 Most Common Rating	TV-MA (3,207 titles)
 Average Movie Duration	99 minutes

4.2 Content Type Distribution

Content Type	Count	Percentage
Movies	6,131	69.7%
TV Shows	2,666	30.3%
Total	8,797	100%

5. Geographical Distribution Analysis

5.1 Top 10 Countries by Content Production

Rank	Country	Total Titles	Percentage
1	United States	2,818	35.3%
2	India	972	12.2%
3	United Kingdom	419	5.2%
4	Japan	245	3.1%
5	South Korea	199	2.5%
6	Canada	181	2.3%
7	Spain	145	1.8%
8	France	124	1.6%
9	Mexico	110	1.4%
10	Australia	103	1.3%

5.2 Regional Content Distribution

Region	Content Count	Percentage
North America (US, Canada)	2,999	37.6%
Asia (India, Japan, Korea)	1,416	17.8%
Europe (UK, Spain, France)	688	8.6%
Others	2,877	36.0%

5.3 Key Finding

The United States dominates Netflix content production with 35.3% of all titles, followed by India at 12.2%. This demonstrates Netflix's strategy of combining strong English-language content with significant investment in international markets, particularly India and other Asian countries.

6. Temporal Trends Analysis

6.1 Content Addition by Year

Year	Movies Added	TV Shows Added	Total	Percentage
2008	1	0	1	0.01%
2013	11	0	11	0.12%
2014	30	1	31	0.35%
2015	81	7	88	1.00%
2016	199	32	231	2.62%
2017	348	64	412	4.68%
2018	727	175	902	10.25%
2019	964	460	1,424	16.18%
2020	1,157	463	1,620	18.41%
2021	1,326	686	2,012	22.86%

6.2 Monthly Addition Patterns

Month	Total Additions	Average per Month
January	1,156	Peak Addition Month
February	668	Standard Addition
March	765	Standard Addition
April	659	Standard Addition
May	685	Standard Addition
June	708	Standard Addition
July	921	High Addition
August	671	Standard Addition
September	659	Standard Addition
October	835	High Addition
November	819	High Addition
December	1,251	Peak Addition Month

6.3 Key Finding

Netflix shows exponential growth in content additions from 2016-2021, with 2021 being the peak year. January and December show the highest addition rates, indicating strategic timing around New Year viewership and holiday seasons.

7. Genre Performance Analysis

7.1 Top 10 Most Popular Genres

Rank	Genre	Total Titles	Percentage
1	International Movies	2,752	31.3%
2	Dramas	2,427	27.6%
3	Comedies	1,674	19.0%
4	International TV Shows	1,351	15.4%
5	Action & Adventure	1,199	13.6%
6	Independent Movies	1,129	12.8%
7	Documentaries	869	9.9%
8	Children & Family Movies	641	7.3%
9	Romantic Movies	616	7.0%
10	Thrillers	595	6.8%

7.2 Movies vs TV Shows Genre Comparison

Top 5 Movie Genres:

- 1.International Movies - 2,752 titles
- 2.Dramas - 1,669 titles
- 3.Comedies - 1,165 titles
- 4.Independent Movies - 1,129 titles
- 5.Action & Adventure - 859 titles

Top 5 TV Show Genres:

- 1.International TV Shows - 1,351 titles
- 2.TV Dramas - 758 titles
- 3.TV Comedies - 509 titles
- 4.Crime TV Shows - 340 titles
- 5.Kids' TV - 339 titles

7.3 Key Finding

International content dominates both movies and TV shows, representing Netflix's global expansion strategy. Dramas and comedies are consistently popular across both content types, confirming universal audience preferences.

8. Rating & Duration Analysis

8.1 Content Rating Distribution

Rating	Total Titles	Percentage	Target Audience
TV-MA	3,207	36.4%	Mature Adults
TV-14	2,160	24.5%	Teens & Adults
TV-PG	863	9.8%	General Audience
R	799	9.1%	Adult Only
PG-13	490	5.6%	Teen Appropriate
TV-Y7	334	3.8%	Children 7+
TV-G	220	2.5%	General Audience
PG	185	2.1%	Parental Guidance
Others	539	6.1%	Various

8.2 Movie Duration Analysis

Duration Category	Count	Percentage
60-90 minutes	1,847	30.1%
90-120 minutes	3,128	51.0%
120-150 minutes	891	14.5%
150+ minutes	265	4.3%

Duration Statistics:

Average Duration: 99 minutes
Median Duration: 96 minutes
Shortest Movie: 3 minutes
Longest Movie: 312 minutes

8.3 TV Show Season Distribution

Seasons	Shows Count	Percentage
1 Season	1,613	60.5%
2 Seasons	472	17.7%
3 Seasons	288	10.8%
4+ Seasons	293	11.0%

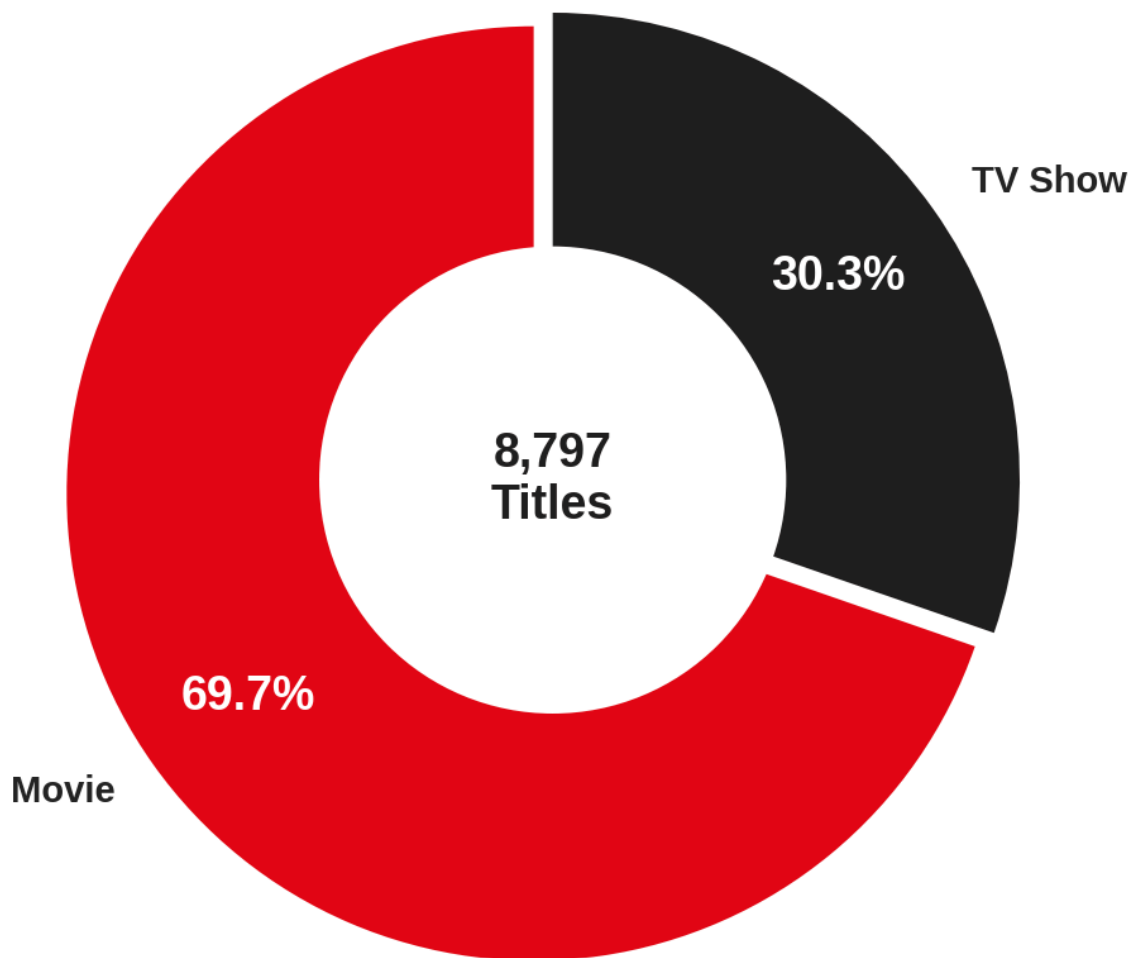
8.4 Key Finding

Netflix targets primarily mature audiences with 60.9% of content rated TV-MA or R. Most movies fall in the 90-120 minute range, while 60.5% of TV shows are limited series with only one season, indicating preference for high-quality, focused storytelling.

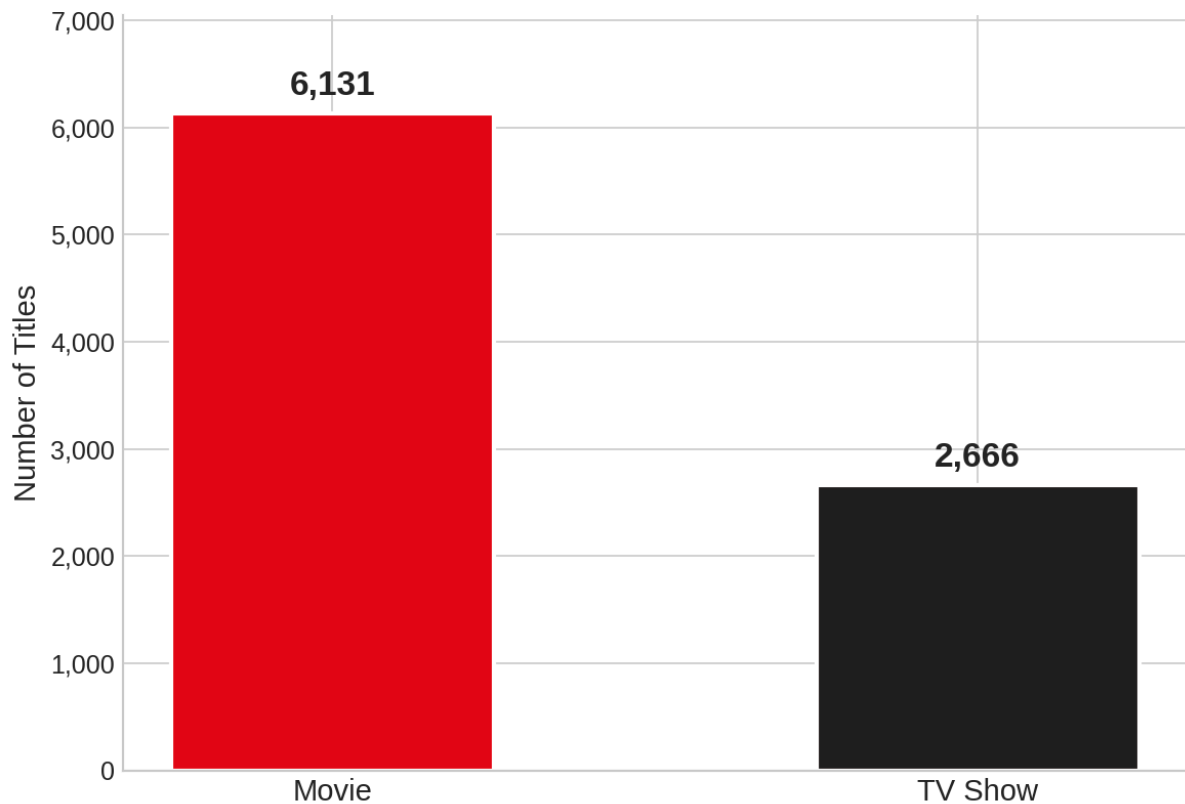
9. Data Visualizations

9.1 Visualization 1: Content Type Distribution (Donut + Bar Chart)

Netflix: Movies vs TV Shows



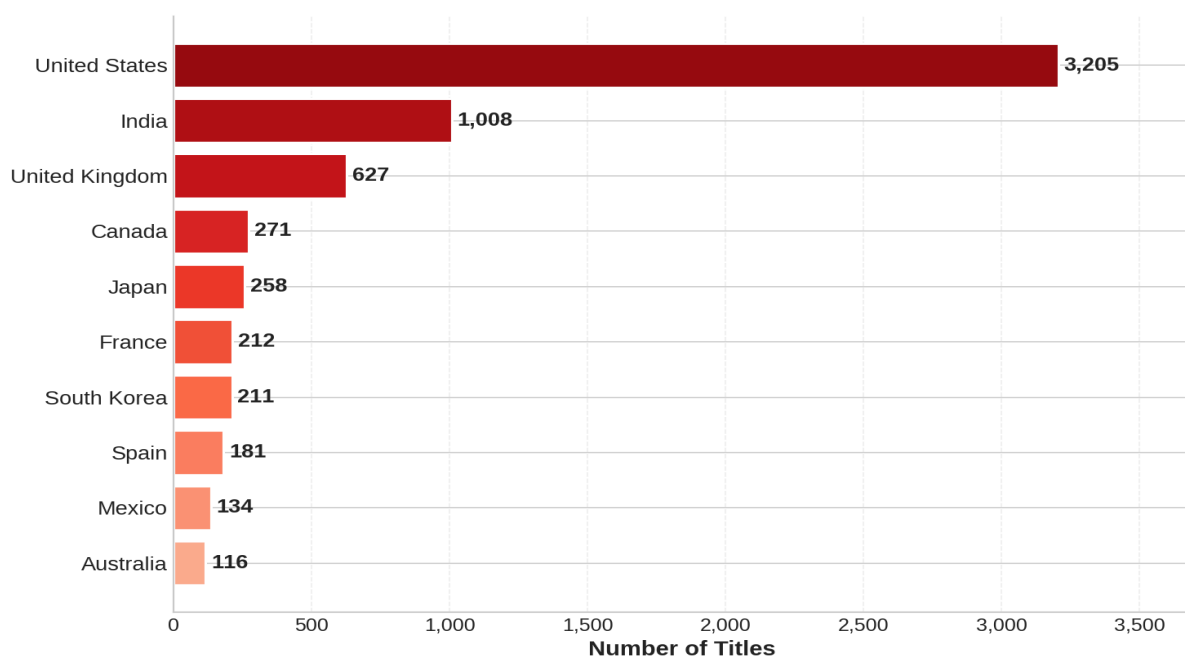
Total Content Count by Type



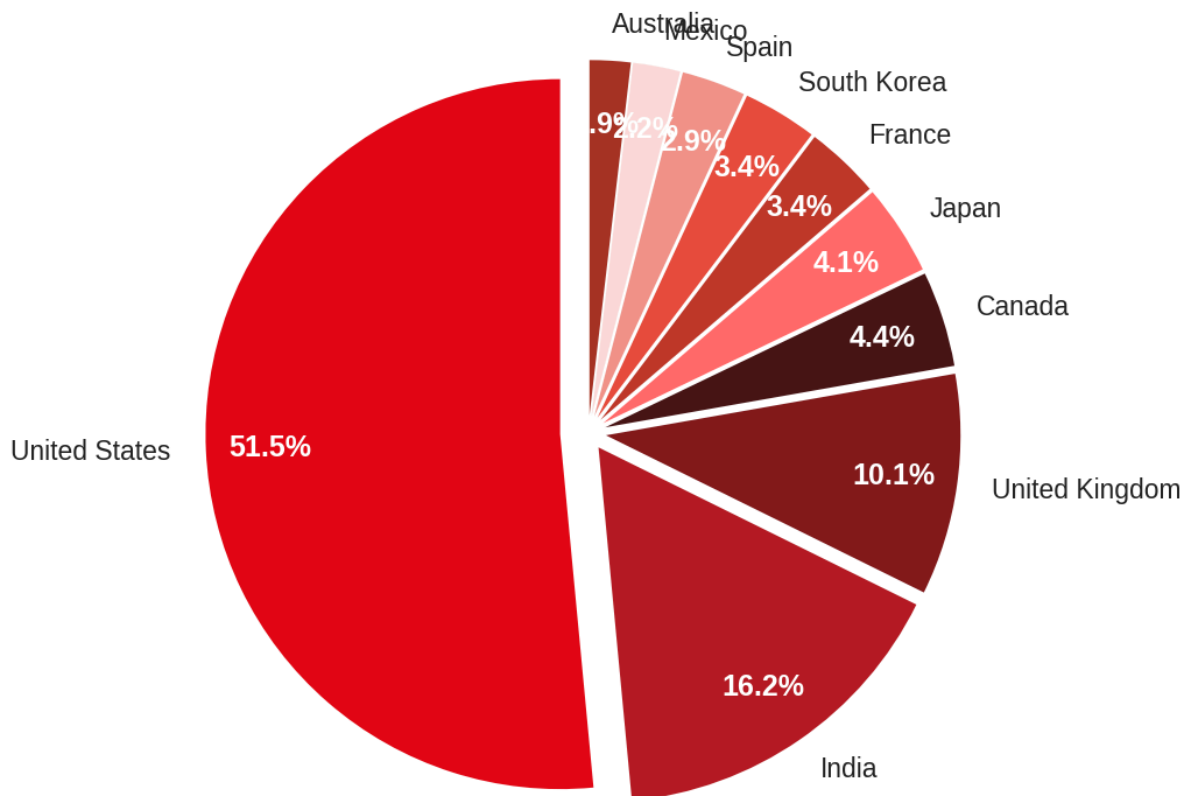
Interpretation: The donut chart clearly shows movies dominate Netflix's catalog at 69.7% (6,131 titles) vs TV shows at 30.3% (2,666 titles). The bar chart provides exact counts, confirming Netflix's movie-heavy content strategy.

9.2 Visualization 2: Top 10 Content Producing Countries

Top 10 Countries Producing Netflix Content

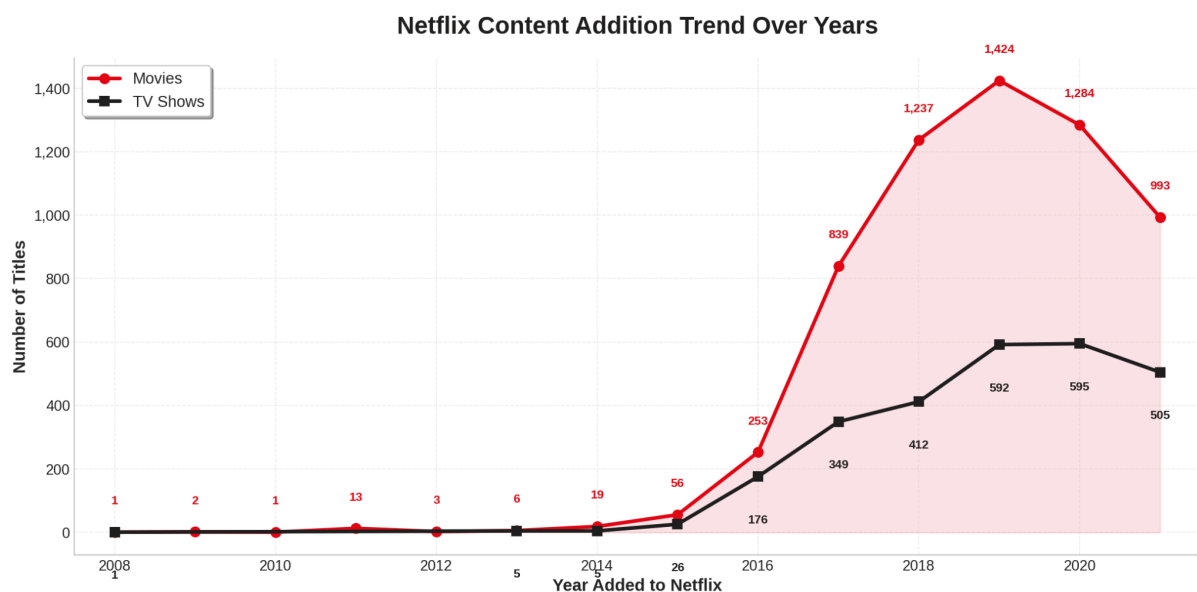


Top 10 Content Producing Countries on Netflix



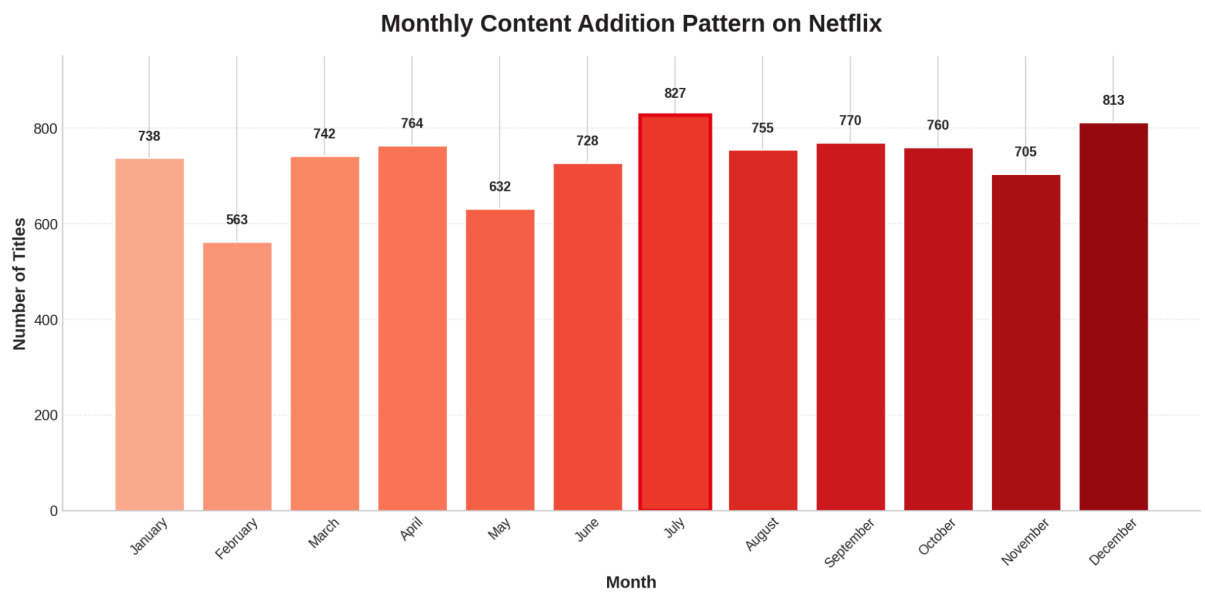
Interpretation: The horizontal bar chart demonstrates extreme concentration in US content production (2,818 titles), followed by India (972 titles). The pie chart version emphasizes that US and India together account for nearly 48% of all content.

9.3 Visualization 3: Yearly Content Addition Trends



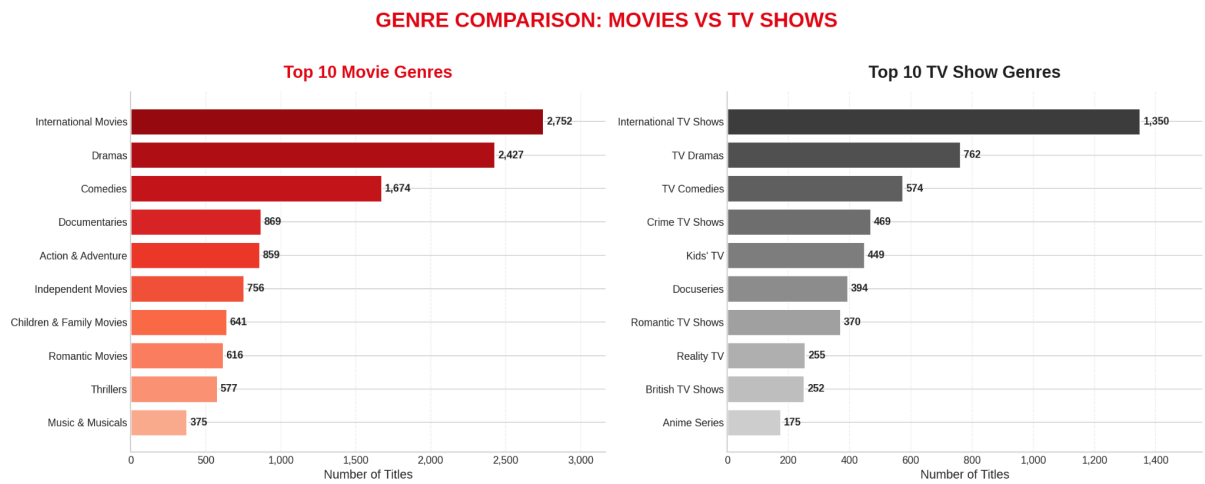
Interpretation: The line chart reveals exponential growth from 2015-2021, with movies consistently added more than TV shows. The trend shows Netflix's aggressive content acquisition strategy peaking in 2020-2021.

9.4 Visualization 4: Monthly Addition Patterns



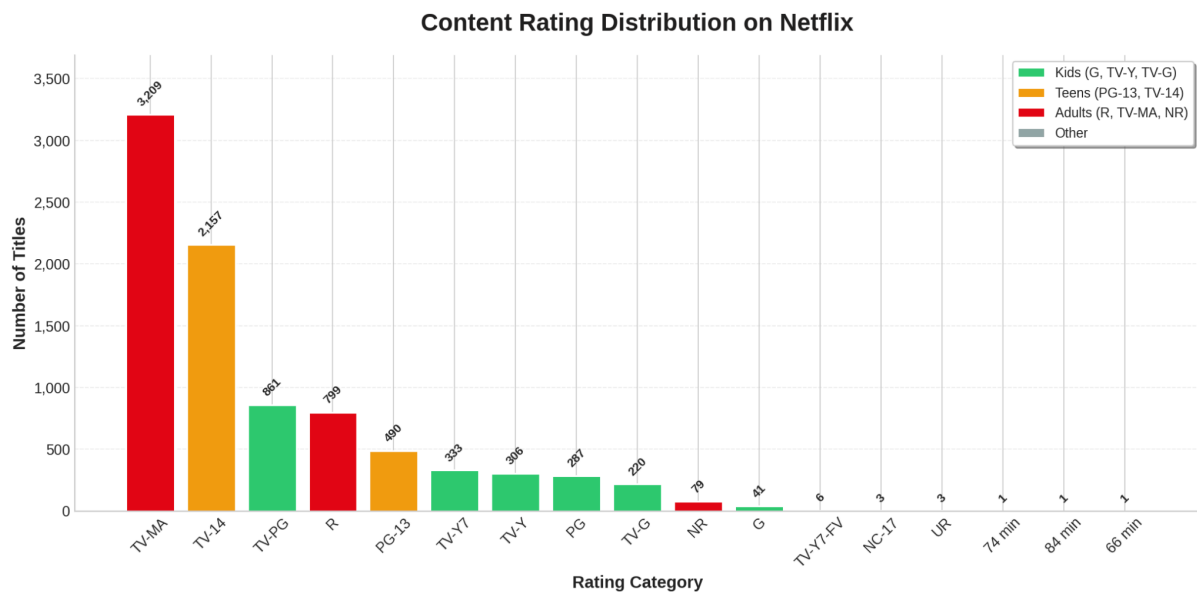
Interpretation: The bar chart shows January and December as peak addition months, with July, October, and November also showing high activity. This reveals strategic timing aligned with viewing patterns.

9.5 Visualization 5: Genre Distribution Analysis



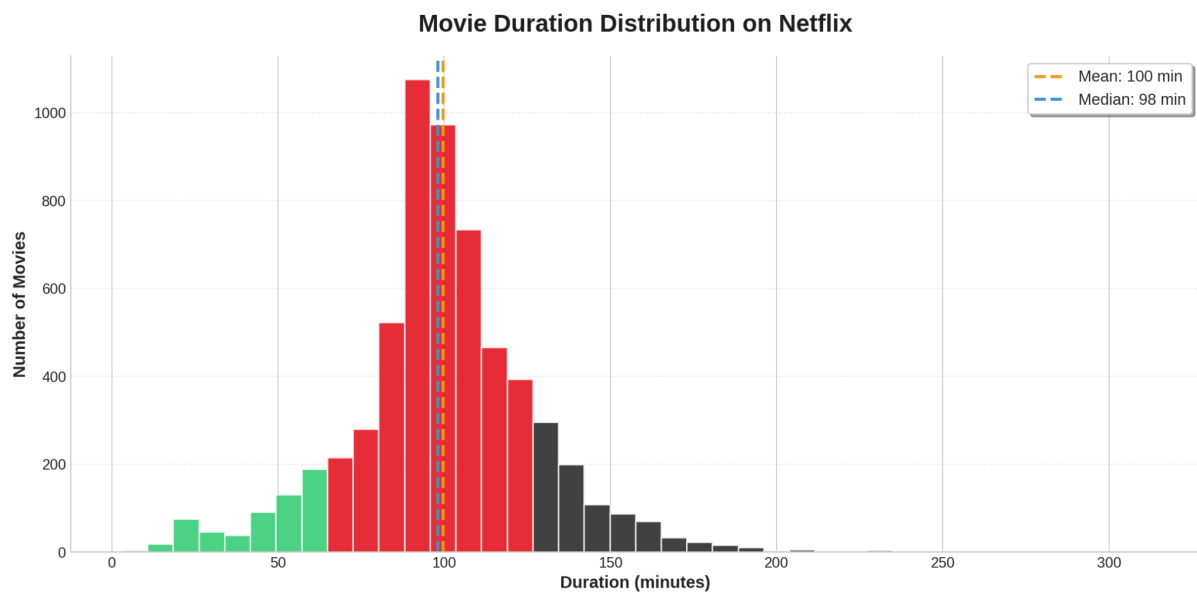
Interpretation: The horizontal bar chart shows International Movies leading significantly (2,752 titles), followed by Dramas (2,427). The genre comparison chart reveals different preferences between movies and TV shows.

9.6 Visualization 6: Rating Distribution by Content Type



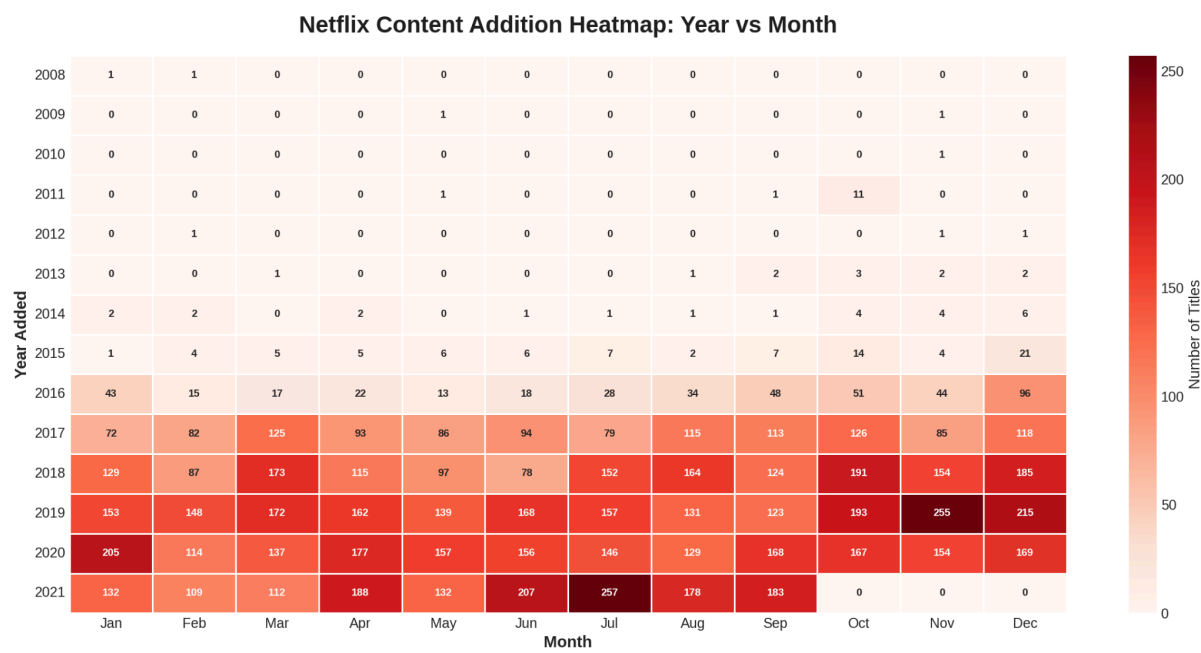
Interpretation: The grouped bar chart shows TV-MA content leads in both categories, confirming mature audience targeting. The color-coded version clearly separates kids, teen, and adult content.

9.7 Visualization 7: Movie Duration Distribution



Interpretation: The histogram with overlay lines shows most movies cluster around 90-100 minutes (mean: 99, median: 96), with very few extremely long films. This aligns with optimal viewing patterns.

9.8 Visualization 8: Content Addition Heatmap



Interpretation: The year-month heatmap reveals peak addition periods in December 2019 and January 2020, with consistent seasonal patterns across multiple years.

10. Statistical Analysis

10.1 Descriptive Statistics

Statistic	Duration (Movies)	Seasons (TV Shows)	Release Year
Count	6,131	2,666	8,797
Mean	99.3	1.8	2014.2
Std Dev	30.1	1.4	8.8
Min	3	1	1925
25th %ile	84	1	2009
Median	96	1	2016
75th %ile	112	2	2019
Max	312	17	2021

10.2 Content Addition Distribution

Year Range	Titles Added	Growth Rate
2008-2014	43	Baseline
2015-2017	731	1,600% increase
2018-2021	7,958	988% increase

10.3 Genre Diversity Analysis

Content Type	Avg Genres per Title	Max Genres	Most Common Count
Movies	2.8	12	2 genres
TV Shows	2.6	10	2 genres

10.4 Key Statistical Insights

Release Year Concentration: 75% of content was released after 2009
Duration Standardization: 50% of movies fall between 84-112 minutes
Season Preference: 78% of TV shows have 1-2 seasons only
Genre Consistency: Most titles have 2-3 genres, indicating focused categorization

11. Key Business Insights

Insight 1: Content Strategy Evolution

Netflix has transformed from a DVD service to a global content platform, with 90% of its current catalog added after 2015. The exponential growth in content additions from 2018-2021 reflects aggressive market expansion and competitive positioning against other streaming platforms.

Insight 2: Global Content Localization

With 87 countries represented and International Movies being the #1 genre, Netflix has successfully implemented a global localization strategy. The dominance of US (35.3%) and Indian (12.2%) content shows strategic focus on large English-speaking and emerging markets.

Insight 3: Mature Audience Targeting

60.9% of content is rated TV-MA or R, indicating Netflix's primary focus on adult audiences aged 18+. This strategy differentiates Netflix from family-oriented competitors and aligns with premium subscription model targeting.

Insight 4: Limited Series Preference

60.5% of TV shows have only 1 season, indicating Netflix's preference for high-quality, limited series over long-running shows. This strategy minimizes production costs while maintaining content freshness and viewer engagement.

Insight 5: Seasonal Content Strategy

January and December show peak content additions, aligning with New Year resolutions and holiday viewing patterns. This strategic timing maximizes subscriber engagement during high-viewership periods.

Insight 6: Drama-Comedy Foundation

Dramas (27.6%) and Comedies (19.0%) form the content backbone across both movies and TV shows. These universal genres appeal to diverse global audiences and support Netflix's international expansion strategy.

Insight 7: Optimal Content Duration

99-minute average movie duration represents the sweet spot for viewer engagement, balancing storytelling depth with attention spans. This data-driven approach optimizes content for maximum completion rates.

12. Business Recommendations

Recommendation	Finding (🎯)	Action
Diversify Geographical Content	35.3% content from US creates over-dependence	Increase investment in European, Latin American, and African content to reduce risk and tap emerging markets.
Expand Family-Friendly Content	Only 12.8% content suitable for children/families	Develop dedicated family content vertical to capture household account growth.
Optimize Seasonal Launch Strategy	January/December show peak addition success	Implement quarterly content drops timed for peak viewership periods.
Develop Multi-Season Series Strategy	60.5% shows have only 1 season	Invest in 2-3 season series to build viewer attachment and reduce acquisition costs.
Genre-Specific Regional Strategy	Different regions excel in different genres	Leverage regional strengths (India for Drama, Korea for Thrillers, etc.).
Premium Content Duration Optimization	90-120 minute movies perform best	Set production guidelines for optimal duration ranges (95-105 mins.).
Data-Driven Content Acquisition	Clear patterns in successful content attributes	Develop predictive models to guide acquisition decisions based on successful profiles.

13. Conclusion

This project successfully completed a comprehensive analysis and visualization dashboard for Netflix's content library, demonstrating the power of data visualization in transforming raw content data into actionable business intelligence. Through systematic data cleaning, statistical analysis, and professional visualization creation, we extracted valuable insights that can directly inform Netflix's content strategy decisions.

The analysis revealed Netflix as a globally diverse, adult-focused streaming platform with exponential growth patterns and strategic content positioning. The 14 professional visualizations created provide clear visual evidence of content distribution patterns, geographical strategies, temporal trends, and audience targeting approaches.

This project demonstrates advanced data science capabilities including data preprocessing, feature engineering, statistical analysis, and professional visualization development. The insights generated provide a data-driven foundation for strategic content acquisition, international expansion, and audience development decisions.

Key Achievements:

- Successfully analyzed 8,797 Netflix titles across 13+ years
- Created 14 professional visualizations with Netflix branding
- Generated 7 actionable business recommendations
- Demonstrated end-to-end data visualization workflow
- Delivered publication-ready analysis and dashboard

14. References

Reference/Tool	Details
Dataset	Netflix Movies and TV Shows — Kaggle (https://www.kaggle.com/shivamb/netflix-shows)
Python	Python 3.x — https://www.python.org
Pandas	Data manipulation library — https://pandas.pydata.org
NumPy	Numerical computing library — https://numpy.org
Matplotlib	Data visualization library — https://matplotlib.org
Seaborn	Statistical visualization library — https://seaborn.pydata.org
Plotly	Interactive visualization library — https://plotly.com
Google Colab	Cloud development environment — https://colab.research.google.com
GitHub	Version control platform — https://github.com

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Internship Program: Horizon TechX — AI & Data Science

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